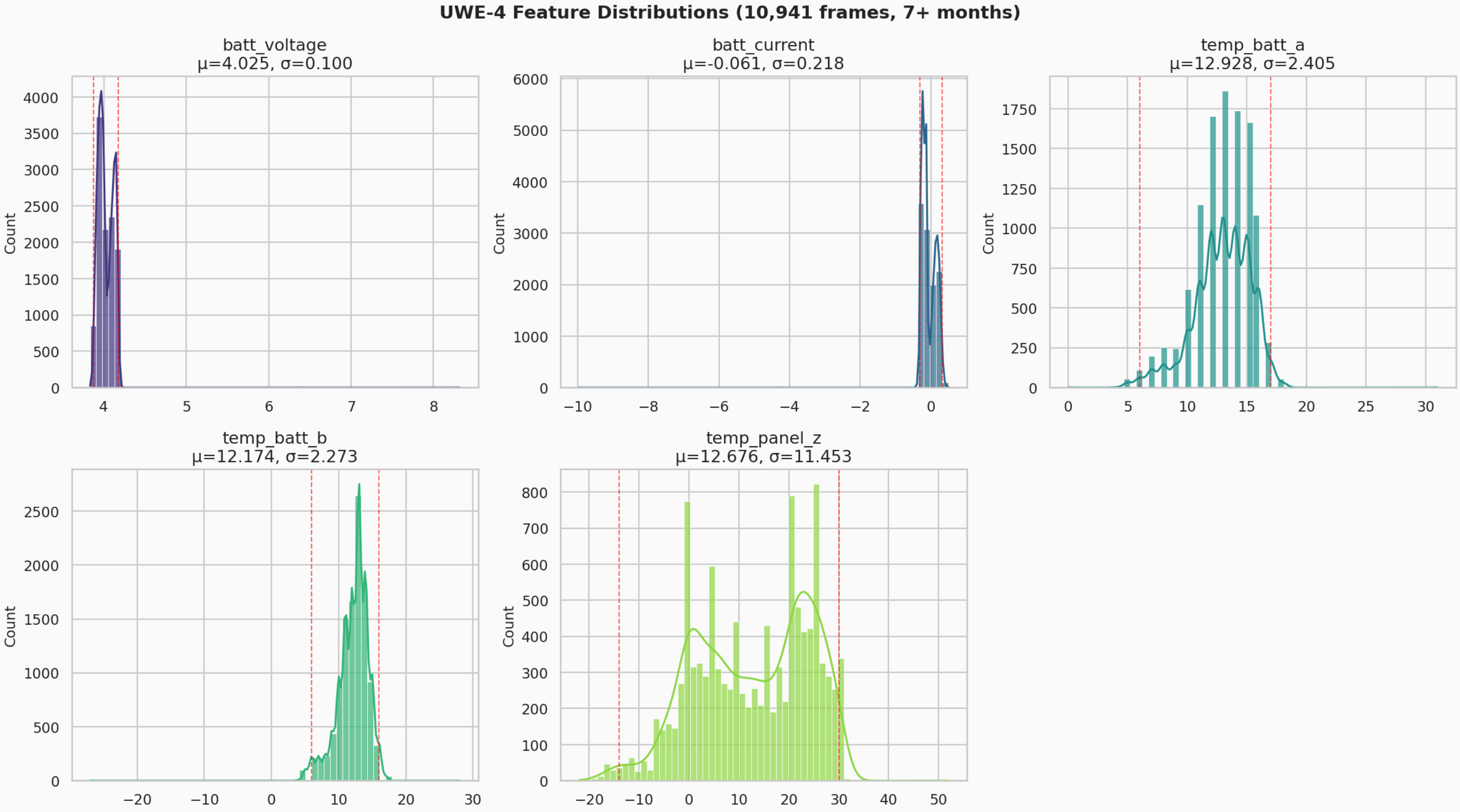
UWE-4 Exploratory Data Analysis

Telemetry Validation, Benchmarking & Dashboard Readiness

1. Data Engineering & Sanity Checks

Pipeline Integrity & Feature Distributions

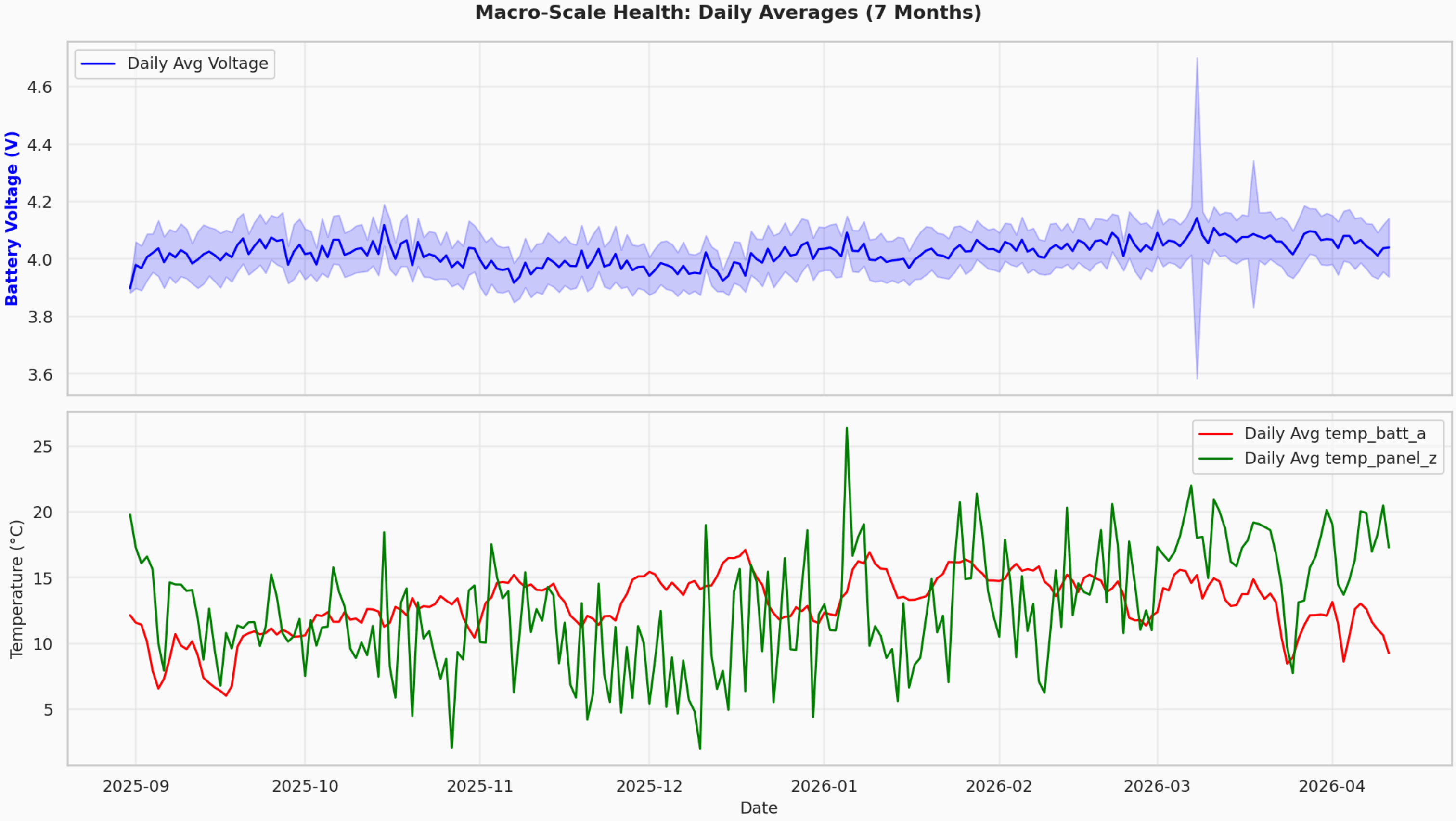
Our core dataset spans 7 months of clean SI-unit data. We actively filter out known physical impossibilities early (e.g. 8V battery spikes) because they are communications errors, not physics anomalies.



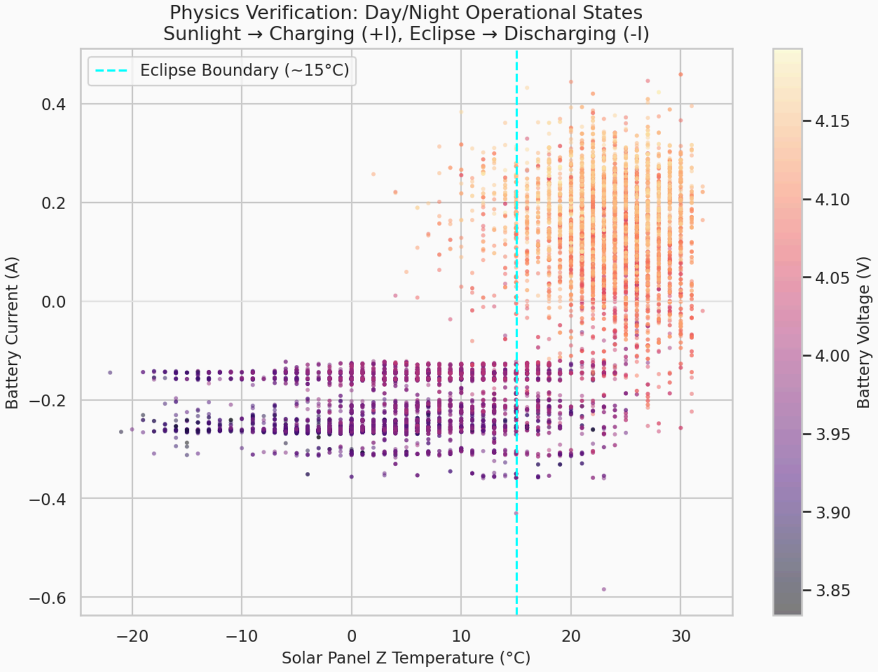
2. Deep-Dive Exploratory Data Analysis

Long-Term Macro Trends (7 Months)

Over the 7-month dataset, we see extreme seasonality in satellite thermodynamics and charge cycles. The baselines shift massively over the year.



The Bimodality Challenge (Day vs. Eclipse)

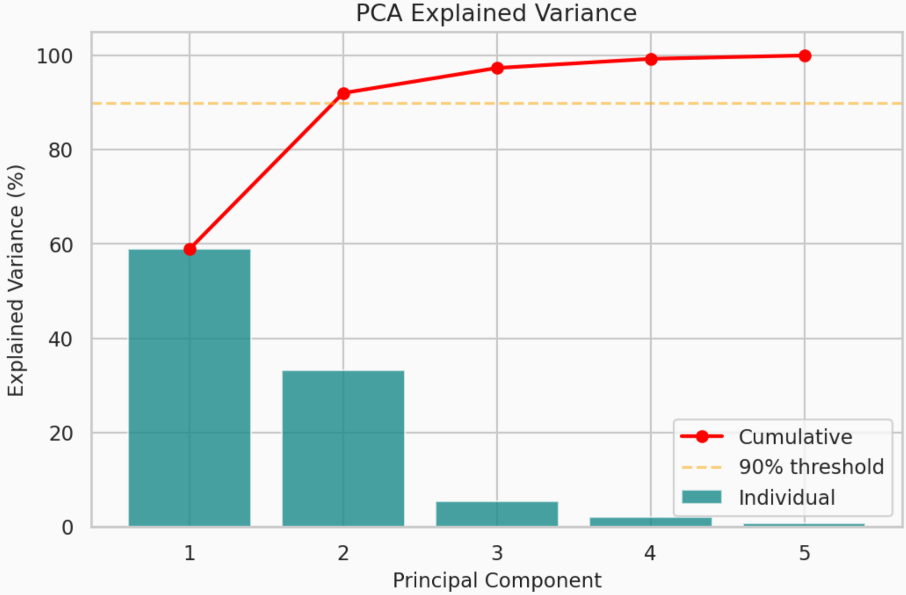


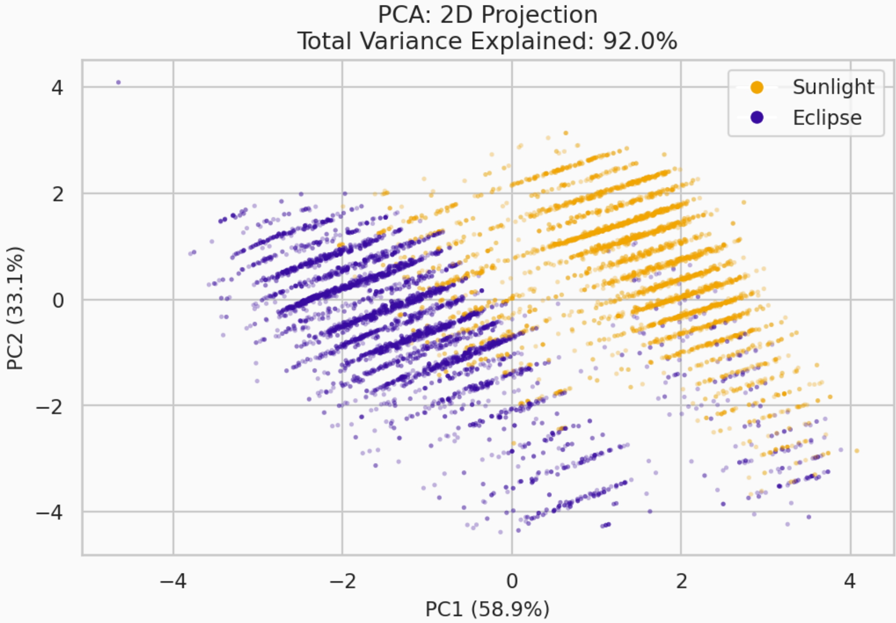
Non-Linear Correlations & States

The transition into eclipse structurally uncouples features. `batt_current` acts differently when `temp_panel_z < 0` (Eclipse) versus when it's `> 20` (Sunlight).

A static threshold logic gate would either trigger 1,000 false positives every orbit, or be so loose that it misses catastrophic component failures.

PCA proves our 5 physical features are heavily correlated and can be mathematically compressed, validating our Autoencoder hypothesis.





3. The Edge Discontinuity

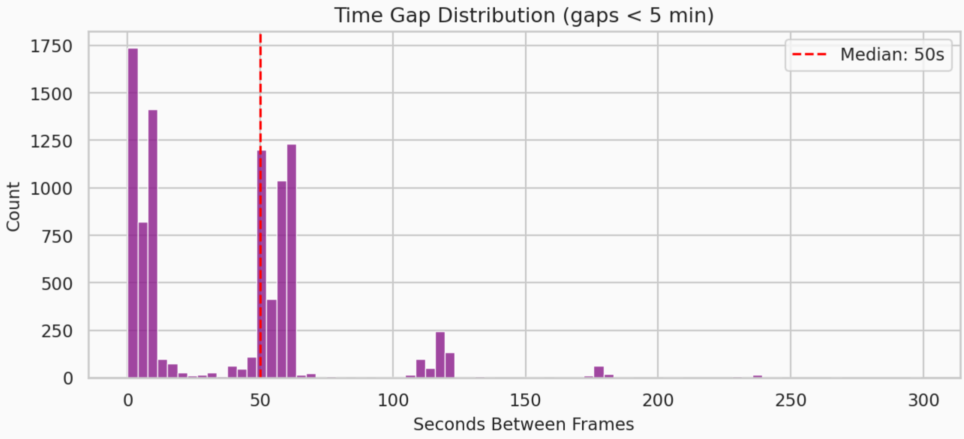
Time Gaps vs Traditional Time-Series

Edge station inference is strictly constrained by Line-Of-Sight passes.

The histogram explicitly visualizes **Intra-Pass** gaps (delays occurring while the satellite is actively overhead).

- **Mode (Peak Frequency):** 0-10s
- **Median (Due to long tails/ clusters):** 39s
- **Gap between passes:** 10 hours! (Filtered from plot)

Why LSTMs / Transformers fail here: Rolling history expires between passes. Attempting to use the frame from



Time Gaps vs Traditional Time-Series

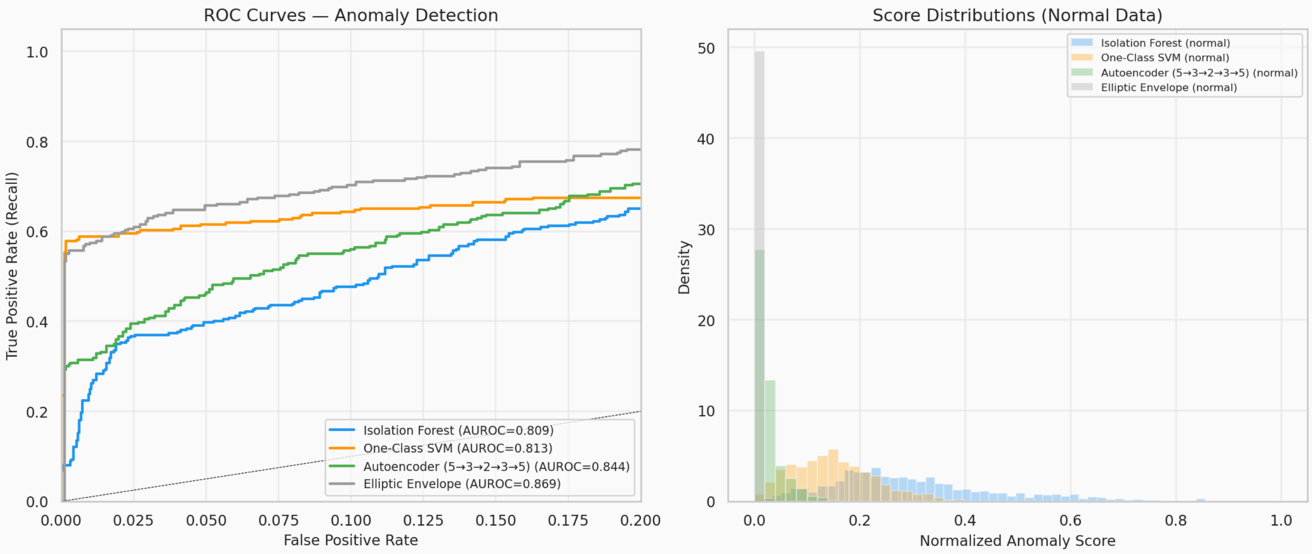
"10 hours ago" breaks physics predictions.

Conclusion: We require **Stateless Ensembles** (evaluating each frame in a vacuum).

4. Model Benchmarking

The Evaluated AI Models

We benchmarked several unsupervised models against synthetic physical faults. The current shipped benchmark script evaluates `Thermal Runaway` and `Panel Failure`; `Sensor Stuck` was an earlier notebook-only experiment.

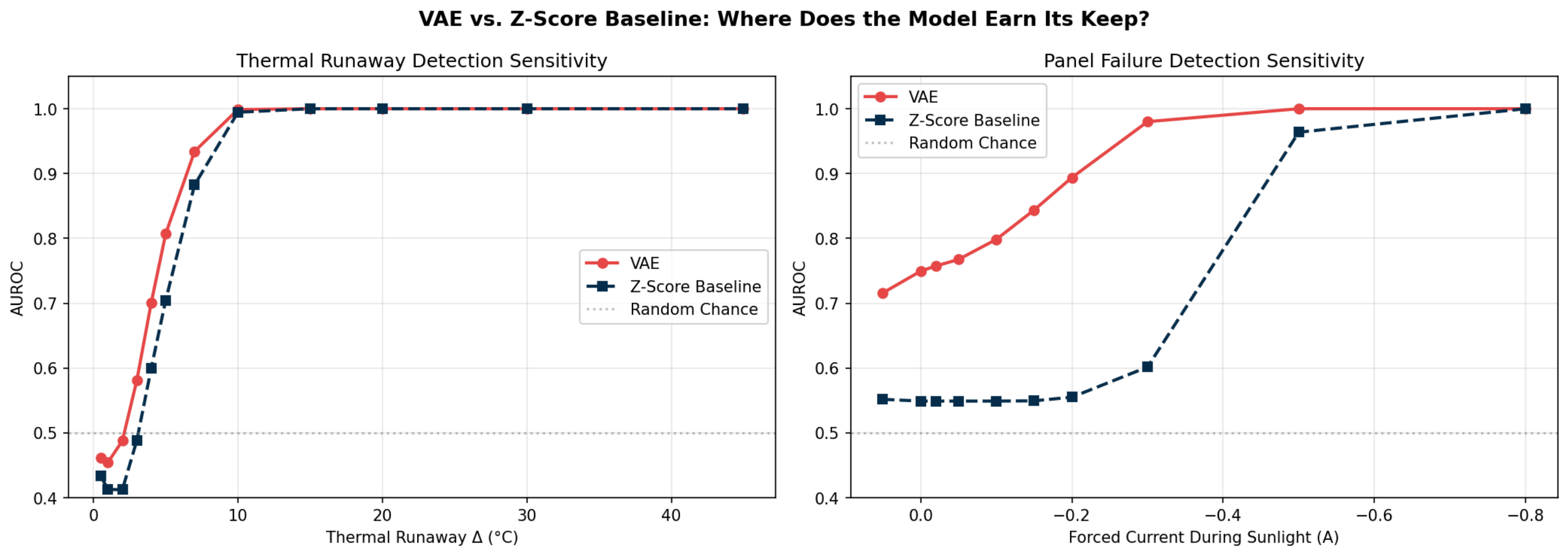


The Standouts:

- **Elliptic Envelope:** Appeared to do well in basic AUROC, but suffered catastrophic 50% Recall failure because its Gaussian boundaries cannot map bimodal clusters (Day/Night) accurately without drawing boundaries too loose.
- **Variational Autoencoder (VAE):** Perfect baseline handling non-linear space with explicit probabilistic modeling.

The “Honest” Benchmark: Sensitivity Sweep

A 100% detection rate on extreme faults proves nothing; basic thresholds can catch +45°C thermal spikes. We swept the fault magnitudes from subtle to extreme to find the operational crossover where the VAE’s multivariate awareness actually outperforms a simple Z-Score limit.



The “Honest” Benchmark: Sensitivity Sweep

The Verdict: The VAE massively outperforms dumb thresholding (Z-Score) during subtle anomalies (like a 0.1A current drop during sunlight), while performing equally well on obvious, extreme faults.

Stage 1: Overall Loss Detector

Context: We dumped the EllipticEnvelope entirely.

Action: Run the VAE. Calculate the sum of its Mean Squared Error (MSE) + Kullback-Leibler Divergence (KLD). In the current repository, the operating threshold is still derived inside offline evaluation rather than persisted during training.

Key Insight: Uses the VAE's intrinsic ability to map non-linear bimodal physics mathematically, but the thresholding path is still benchmark-only.

Stage 2: Per-Node Diagnoser

Context: We must isolate what broke during an anomaly.

Action: Inspect the Mean Squared Error node-by-node. The feature with the highest individual MSE represents the hardware failure.

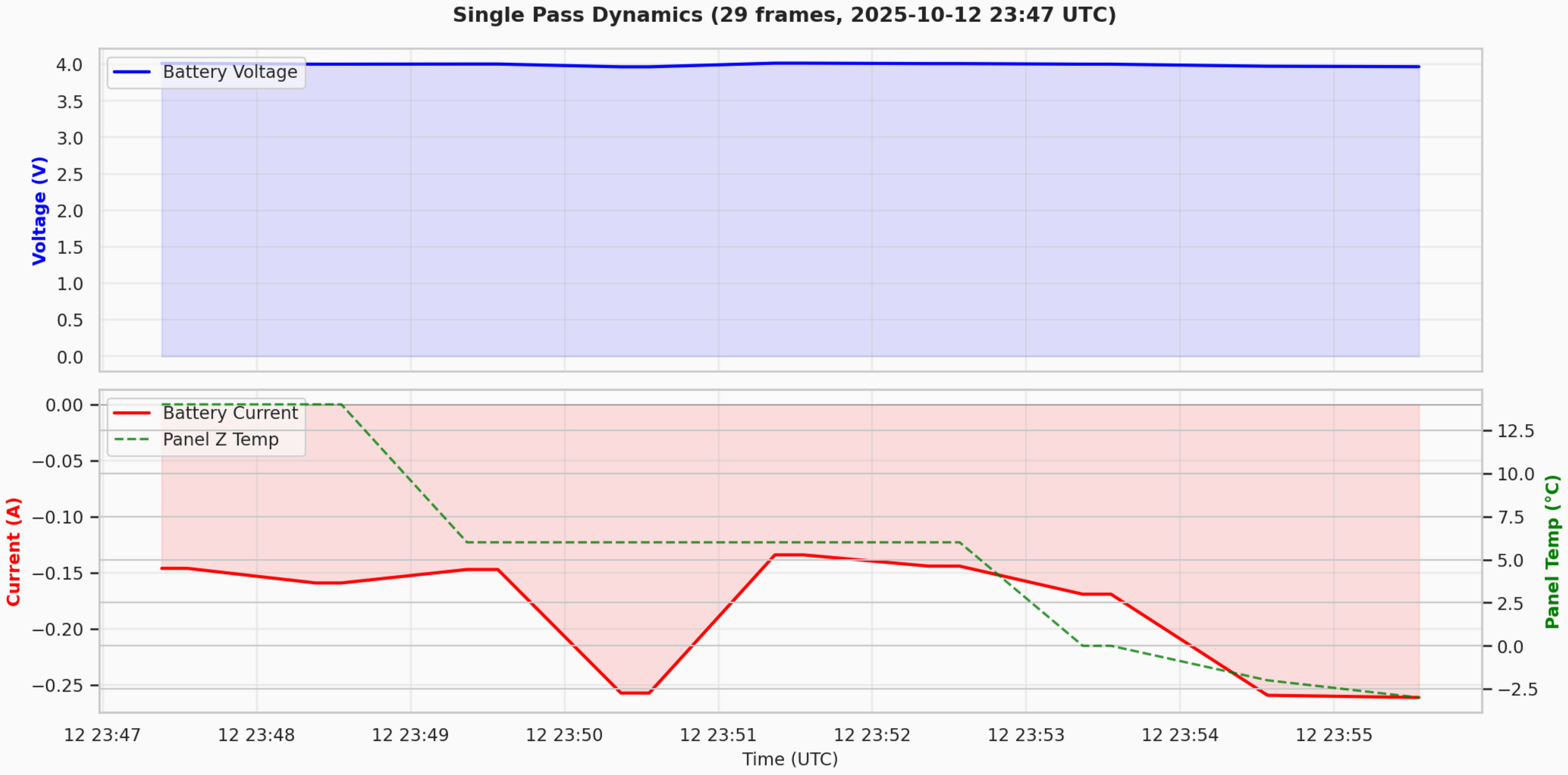
Key Insight: Provides offline fault-localization signal in the benchmark flow.

5. Dashboard Blueprints

Micro Scale Pass Dynamics (Widget 1)

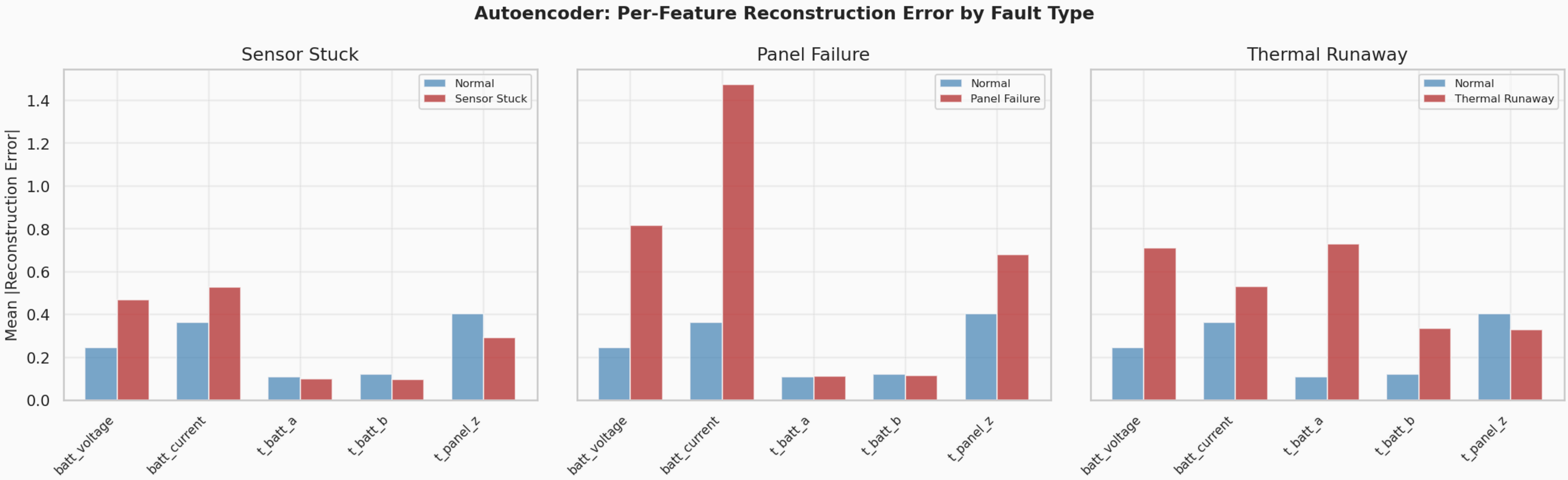
Planned dashboard widget: a time-series view tracking variables throughout a specific satellite pass overhead.

Micro Scale Pass Dynamics (Widget 1)



Autoencoder Feature Imputation (Widget 2)

When Stage 2 triggers, the UI will display **Actual Values** against the Autoencoder's **Reconstructed/ Expected Values** to show the exact delta forcing the anomaly.



Final Live Dashboard Strategy

To productize “The Watchdog”, we would transition these findings into a unified `marimo` reactive notebook or service-backed UI. This dashboard is not yet implemented in the repository.

Key Dashboard Widgets to build:

1. **Live Pass View:** Telemetry graphing in real-time as packets decode.
2. **Gauge Cluster:** Real-time dials mapping values to historical reference bounds.
3. **Anomaly Probability Dial:** Stage 1 anomaly score mapped from 0-100%.
4. **Subsystem Blame Chart:** Stage 2 `Autoencoder` feature-reconstruction radar chart.